TISSUE REPAIR & RECOVERY

GHK-Cu — the full review

COPPER-BINDING TRIPEPTIDE

This is the deeper look behind the short version — what the copper actually does, what the topical trials measured, why the injectable form is a different question, common misunderstandings, who it is not for in full, and what your consultation covers.

What it is

GHK is a very small natural molecule — three amino acids — that circulates in human blood and is released from tissue after an injury. It binds copper, and that copper-carrying form is called GHK-Cu.

It has been used in skin care for decades and is best known there. More recently, it is used by people looking at tissue repair, skin quality, and recovery, in topical and injectable forms.

What people use it for

USED FOR	EVIDENCE	WHAT THAT MEANS
Fine lines, firmness and skin density	**Limited human data**	Small controlled trials of the topical form measured changes in skin appearance.
Collagen production in skin	**Limited human data**	The best-supported mechanism, again from topical studies.
Wound healing and scar reduction	**Animal data only**	Studied in animal models; no human trial has tested it for this.

How it works

GHK-Cu's activity is tied to its role as a copper carrier: copper is a cofactor for several enzymes involved in collagen and elastin production and in antioxidant defense. GHK naturally rises at sites of tissue injury, which is part of why it has been studied as a signal involved in wound healing.

In skin, the proposed mechanism is stimulation of collagen synthesis and modulation of the skin's structural proteins, which is the basis for its use in cosmetic formulations aimed at fine lines and skin firmness.

What the research actually looked at

Small controlled trials of the topical form have measured changes in skin appearance — fine lines, firmness, and skin density — making topical use the best-supported application of GHK-Cu. Collagen production in skin is considered the most solidly supported mechanism, again based on topical studies.

Wound healing and scar reduction have been studied in animal models but not in human trials. The injectable form used in some clinical settings has a substantially thinner evidence base than the topical form, and evidence for one route should not be assumed to transfer to the other.

Common misunderstandings

- **“The skin evidence means the injectable form works too.”** Evidence for a compound applied one way does not automatically transfer to a different route; the injectable form has far less human data than the topical form.
- **“It's proven for wound healing in people.”** Wound healing and scar reduction have been studied in animal models; no human trial has tested this use.
- **“More copper is always better.”** Copper handling is tightly regulated by the body, and anyone with a copper-handling disorder or already taking copper supplements should not add GHK-Cu without medical input.
- **“It's a new discovery.”** GHK is a naturally circulating molecule that has been used in topical skin-care formulations for decades.

Who it is not for

- Anyone with Wilson disease or another copper-handling disorder, or anyone already taking copper supplements.
- Anyone pregnant or breastfeeding.
- Anyone with an active cancer diagnosis, without discussing it first.
- Anyone with sensitive skin, eczema or rosacea should patch-test any topical first.

Competitive athletes: GHK-Cu is not currently named on the WADA Prohibited List. Athletes in tested sport should still confirm current status with their sport's anti-doping authority before use, particularly for compounded injectable forms, given the contamination risk inherent to compounded products.

Who makes this, and under what rules

Everything dispensed here is prepared by a licensed 503A compounding pharmacy — not bought from a supplier and repackaged, and not sourced from an online seller.

That designation carries specific, enforceable obligations:

- **A prescription in your name.** Compounded from a prescription for one identified patient, after an evaluation.
- **Licensed and inspected.** A state licence, subject to inspection by the state board of pharmacy.
- **Ingredients that can be traced.** A certificate of analysis from an FDA-registered facility, back to a named source and lot.
- **Sterile preparation to a published standard.** USP standards.
- **A paper trail.** A lot number and beyond-use date on every preparation.

A compounded preparation is not an FDA-approved medicine — but it is the difference between a licensed pharmacy working to a standard and an unregulated website.

Where this stands

Never approved as an injectable medicine. Appears in approved cosmetic products applied to skin; the injectable form used in clinics is not an approved medicine.

The topical evidence is better than the injectable evidence.

Evidence for a compound applied one way does not transfer to a different route.

Clinical detail

GHK is a naturally occurring tripeptide (glycyl-L-histidyl-L-lysine) found in human plasma; GHK-Cu is its copper-bound form.

Levels of natural GHK in blood are reported to decline with age.

Used in cosmetic formulations for decades; several topical products containing GHK-Cu are marketed as cosmetics, not as drugs.

Small controlled human trials of the topical form have measured skin firmness, density and fine lines.

Wound-healing and scar-reduction data comes exclusively from animal models.

The information on this page is general education. It is not medical advice and it is not a recommendation for any treatment.

GHK-Cu is not an FDA-approved medicine for the uses described here. A compounded preparation is not an FDA-approved product and has not been reviewed by the FDA for safety, effectiveness or quality.

Descriptions of what this is used for reflect how it is commonly used and discussed. They are not claims that it is effective for any purpose.

Any prescription follows an individual evaluation by a licensed provider and is issued only where that provider judges it clinically appropriate. A prescription is never guaranteed.